

Attention Beyond the Training Length

How Score Scaling Controls Length Generalization in a Reduced Binary Attention Classifier

Aaron Shin

Deep Network Understanding Lab
Dickinson College

Motivation

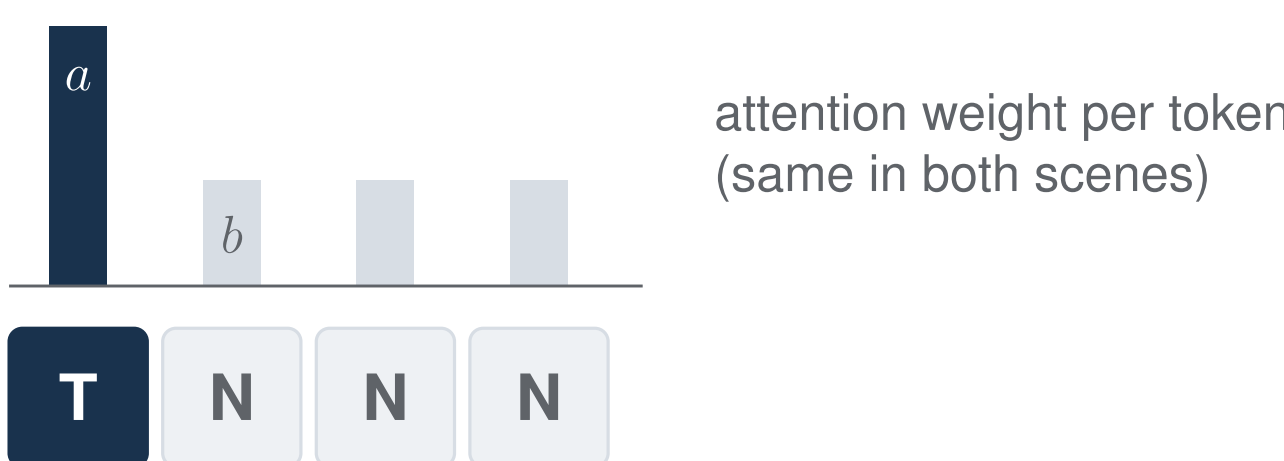
Training on short sequences does not guarantee reliable performance on much longer inputs [2]. As sequence length grows, the target must compete with more non-target tokens for the same attention budget.

How must attention sharpen with length to remain concentrated on the target?

Why Fixed Attention Dilutes with Length

The model detects whether a designated target token is present in the sequence.

Short sequence



Long sequence



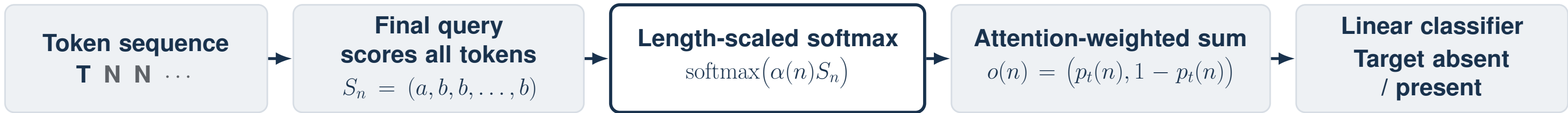
(schematic)

A fixed score margin separates the target from each non-target, but cannot offset their growing aggregate weight.

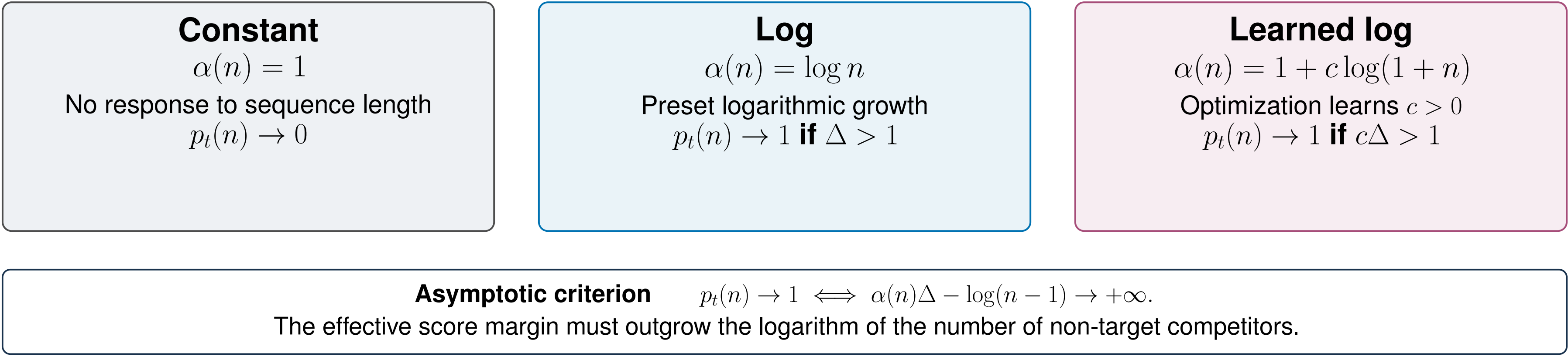
Let $p_t(n)$ be the target's share of attention. With target score a and $n - 1$ identical non-target scores b , standard softmax attention [1] gives

$$p_t(n) = \frac{1}{1 + (n - 1)e^{-\Delta}} \rightarrow 0 \text{ as } n \rightarrow \infty, \quad \Delta = a - b > 0.$$

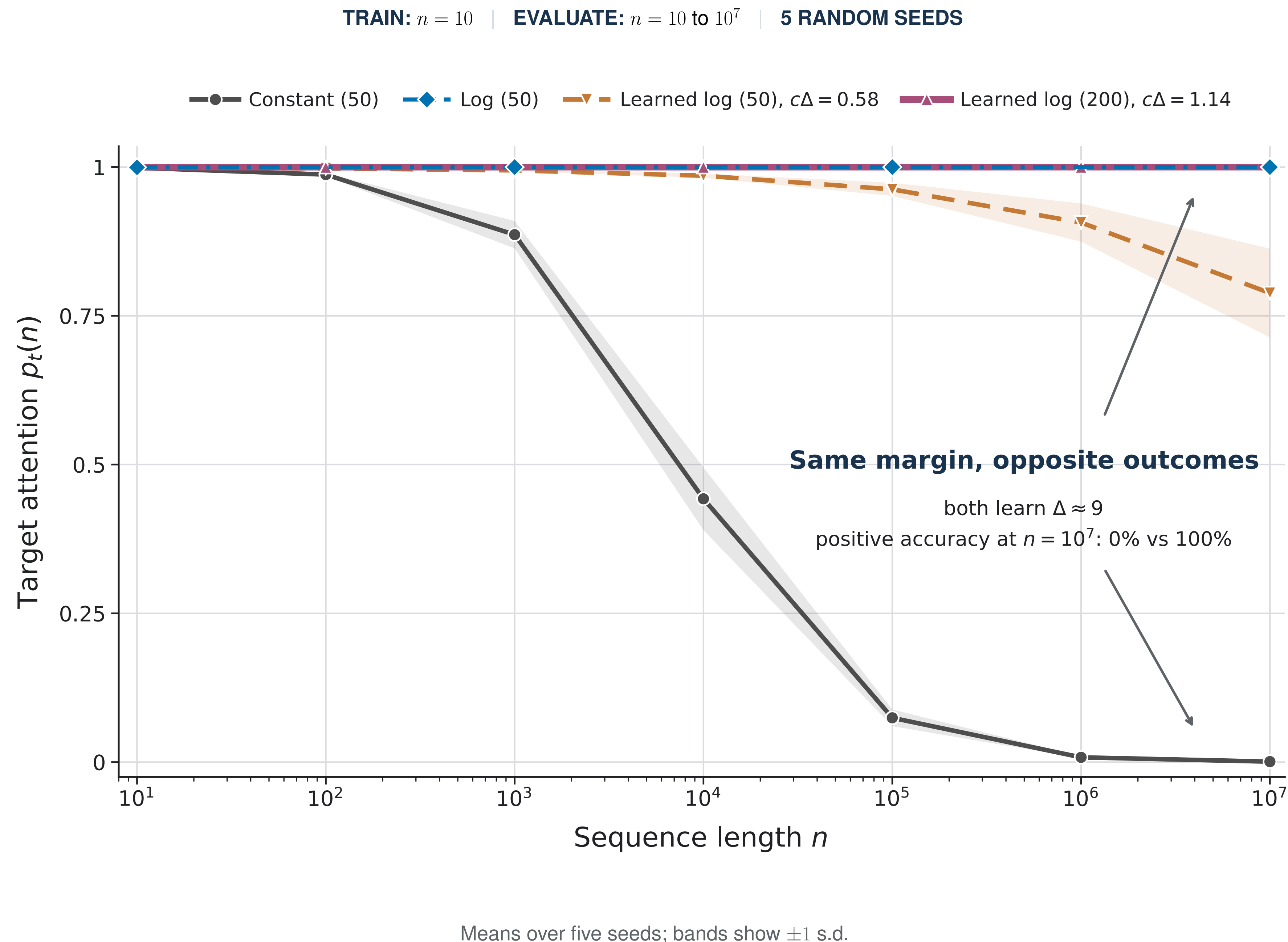
How Length-Aware Scaling Counters Dilution



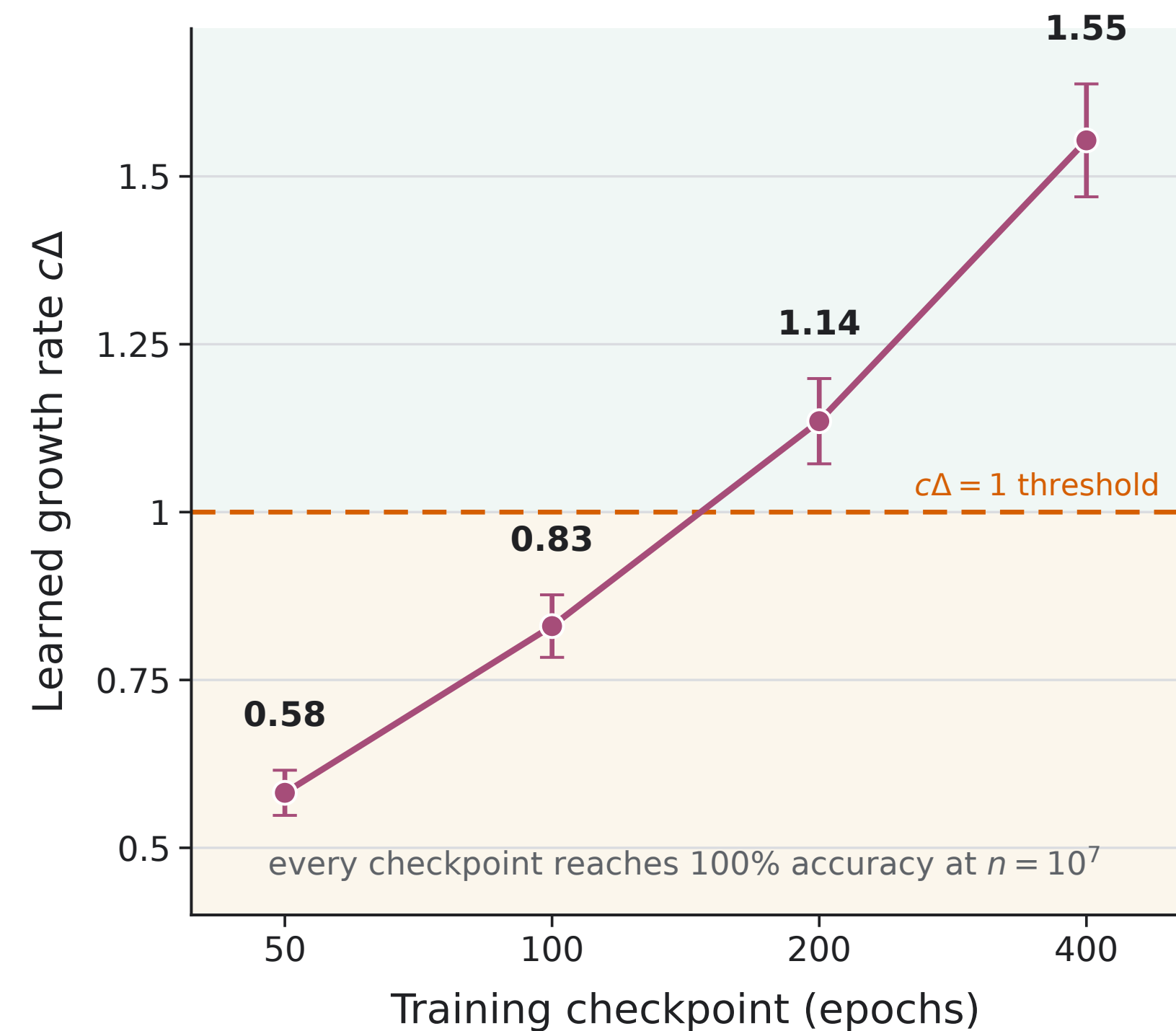
Before softmax, every score is multiplied by a positive factor $\alpha(n)$: the ranking is unchanged, but the effective score margin grows from Δ to $\alpha(n)\Delta$.



Same Training-Length Accuracy. Sharply Different Long-Length Behavior.



Learned-Log Threshold



Means over five seeds; error bars show ± 1 s.d.

A Finite Pass Can Hide Later Failure. Checkpoints below $c\Delta = 1$ pass the $n = 10^7$ benchmark yet are predicted to fail at greater lengths.

Key Takeaways & Scope

1 Fixed scaling postpones, but cannot prevent, target-attention dilution.

2 Sufficiently strong logarithmic sharpening can prevent dilution.

Target attention converges to one when $\Delta > 1$ (Log) or $c\Delta > 1$ (Learned log).

Scope. These thresholds are exact for the reduced two-score classifier studied here; whether analogous conditions hold in full transformers remains open.

References & Acknowledgments

[1] Vaswani et al. (2017). *Attention Is All You Need*. NeurIPS.
[2] Press et al. (2022). *Train Short, Test Long: Attention with Linear Biases Enables Input Length Extrapolation*. ICLR.

Acknowledgments. I thank Professor MacCormick for proposing the reduced-model direction and for guidance on the scope and presentation of this work.